ENCRYPTED DM–TMA

User Tutorial for the Signal-Encrypting DM–TMA Demonstrator

Offline web application guide ů version 1.0

Supporting document for the draft paper

Time-Modulated Arrays with Signal Encryption for Enhanced Directional Modulation Security

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September 18, 2026

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1 Purpose of this tutorial

This document explains how to open, operate, and interpret the *Signal-Encrypting DM–TMA Demonstrator*. The application interactively reproduces the analytical model of the associated paper: a time-modulated uniform linear array (TMA), with two dominant harmonics, independently steerable beams, complementary sum/difference patterns (Σ/Δ), pseudorandom delay updates, and coherent key-based recovery.

The guide is intended both for a first demonstration and for technical analysis. In particular, it details:

- the physical meaning of all sliders, fields, and buttons;
- which quantities are recalculated when each parameter is modified;
- how to read the patterns, link metrics, and constellations;
- how to reproduce the numerical case from the paper;
- how to export results and perform comparative experiments;
- the assumptions and limitations of the model.

Key idea

The demonstrator is not a precomputed animation. Every control change locally recalculates the delays, array factors, angular metrics and, when applicable, the QPSK constellations. No data are transmitted over the Internet.

2 Getting started

2.1 Direct opening

1. Fully extract the archive `TMA_Signal_Encryption_Demo_Web.zip`.
2. Open the folder `TMA_Signal_Encryption_Demo_Web`.
3. Double-click `index.html`.

The default browser will open the application. Python, MATLAB, Node.js, a ChatGPT account, and an Internet connection are not required. Alternatively, the folder can be opened in Visual Studio Code and run with *Live Server*; this step is optional.

2.2 Included files

File	Function
`index.html`	Entry point; supports direct opening.
`app.js`	Self-contained interface and numerical engine.
`styles.css`	Visual presentation and responsive layout.
`data/model-info.json`	Metadata, implemented equations, assumptions, and benchmark.
`README_ES.txt`	Brief opening and scope instructions.
`MANIFEST.json`	File sizes and SHA-256 hashes for checking distribution integrity.

If the browser displays a blank page

Check that the ZIP archive has been fully extracted and that `index.html`, `app.js`, and `styles.css` remain in the same folder. If a corporate policy prevents JavaScript from running from a local file, use Visual Studio Code's *Open with Live Server* option.

3 Minimum model background

3.1 Array and useful harmonics

The system uses $2N$ isotropic elements with uniform excitation and spacing d expressed in carrier wavelengths. Each element receives the premodulated signal $s(t)$ through a cell with four delays:

$$h_n(t) = \frac{1}{\sqrt{2}} a\big(t - D_n^1\big)\, b\big(t - D_n^3\big) + \frac{1}{\sqrt{2}} a\big(t - D_n^2\big)\, b\big(t - D_n^4\big)\,. \tag{1}$$

Delays D_n^1 and D_n^3 synthesize the sum-type (Σ) patterns, whereas D_n^2 and D_n^4 synthesize the difference-type (Δ) patterns. The demonstrator focuses on the two dominant harmonics

$$f^{-1,1} = f_c - f_a + f_b, \qquad f^{1,1} = f_c + f_a + f_b, \tag{2}$$

which can be independently steered toward θ_2 and θ_1, respectively.

3.2 Bandwidth and pseudorandom updating

For QPSK with root-raised-cosine filtering, the demonstrator computes

$$B_f = \frac{R_b(1+\alpha)}{2}, \qquad f_b = \rho B_f, \qquad f_a = \frac{f_b}{2}, \tag{3}$$

where $\rho = f_b/B_f$ is the control labeled `Spectral spacing`. The delay update interval is

$$T_{\mathrm{PR}} = \zeta T_b = \frac{\zeta}{f_b}. \tag{4}$$

In each interval, two independent pseudorandom phases, ϕ_Σ and ϕ_Δ, are selected from an alphabet of M equiprobable values:

$$\phi_\Sigma, \phi_\Delta \in \left\{ \frac{2\pi k}{M} \right\}_{k=0}^{M-1}. \tag{5}$$

An authorized receiver knows the corresponding sequence and can remove the pseudorandom phase. Recovery remains direction-dependent because the Σ and Δ contributions interfere spatially.

3.3 Displayed metrics

For the selected harmonic, the application evaluates the SINR of the two keys from the normalized pattern powers:

$$\gamma_{\Sigma}(\theta) = \frac{|F_{\Sigma}(\theta)|^2}{|F_{\Delta}(\theta)|^2 + P_N}, \tag{6}$$

$$\gamma_{\Delta}(\theta) = \frac{|F_{\Delta}(\theta)|^2}{|F_{\Sigma}(\theta)|^2 + P_N}, \tag{7}$$

with $P_N = 10^{-\mathrm{SNR}/10}$. The displayed secrecy capacity is

$$C_s = \left[\log_2(1+\gamma_b) - \log_2(1+\gamma_e)\right]^+, \tag{8}$$

where γ_b corresponds to the intended direction and γ_e to the most favorable unintended point outside a $\pm 8^\circ$ sector around the main beam.

The BER estimate uses the coherent QPSK expression. In no-key mode, $\mathrm{BER} \simeq 0.5$ is displayed, corresponding to random decisions.

4 Interface organization

The window is divided into four areas:

1. **Header and global actions**: title, offline status, and export buttons.
2. **Derived-quantity strip**: B_f, the frequencies of the two harmonics, T_{PR}, and ideal efficiency.
3. **Left panel**: global parameters for the array, signal, and delay quantization.
4. **Main area**: three tabs for beams/spectrum, encrypted link, and delay synthesis.

4.1 Using the sliders

Each slider displays its current value at the upper right and the allowed endpoints below the track. It can be dragged with the mouse or focused with a click and adjusted with the arrow keys. Each keystroke changes exactly the increment specified in the tables in this tutorial.

The `Array & beams`, `Signal`, and `Hardware realism` headings are collapsible panels. Clicking a heading only shows or hides the controls; it does not modify their values.

5 Left panel: adjustable parameters

5.1 `Array & beams` group

Table 1: Geometry and steering controls.

Control	Range; step	Paper	Meaning and effect	Purpose
`Elements (2N)`	8–40; 2	20	Total number of elements. Only even values are accepted because the Δ synthesis divides the array into two halves. Increasing $2N$ enlarges the aperture, narrows the main beam, and changes the sidelobe structure.	Study angular resolution, ideal array gain, and spatial contrast.
`Spacing`	0.25–$0.75\lambda_c$; $0.01\lambda_c$	$0.50\lambda_c$	Normalized uniform spacing. It changes the physical aperture and the pattern. Above $0.5\lambda_c$, especially for large scan angles, grating lobes may appear.	Analyze trade-offs among size, beamwidth, and angular ambiguities.
f_+ `beam` θ_1	10–170°; 1°	110°	Direction of the Σ maximum of harmonic $\omega^{1,1}$ (cyan family). Its complementary Δ pattern places a null in the same direction.	Demonstrate steering of the first beam without moving the second.
f_- `beam` θ_2	10–170°; 1°	60°	Direction of the Σ maximum of harmonic $\omega^{-1,1}$ (orange family). The associated Δ pattern places its null there.	Demonstrate independent steering of the second beam.

The angle θ is measured from the positive array axis: 90° is *broadside*, while $0^\circ/180^\circ$ are the two *endfire* directions. The model uses $\cos\theta$ in the spatial phase.

5.2 `Signal` group

Table 2: Signal and update controls.

Control	Range; step	Paper	Meaning and effect	Purpose
`Carrier`	1–$12\,\mathrm{GHz}$; $0.05\,\mathrm{GHz}$	$2.4\,\mathrm{GHz}$	Carrier frequency f_c. It sets $\lambda_c = c/f_c$, the absolute harmonic frequencies, and the aperture expressed in centimeters. Since d is specified in λ_c, it does not by itself alter the normalized angular pattern.	Scale the same normalized design to other frequency bands.
`QPSK bit rate`	0.5–$20\,\mathrm{Mbit/s}$; $0.5\,\mathrm{Mbit/s}$	$2\,\mathrm{Mbit/s}$	Bit rate R_b. Through Eq. (3), it changes B_f and therefore f_a, f_b, T_{PR}, and the physical delay values.	Evaluate temporal scalability and hardware requirements.
`RRC roll-off`	0–0.50; 0.05	0.25	Root-raised-cosine roll-off factor α. Increasing it broadens B_f; with f_b/B_f held constant, it also increases the absolute spacing of the replicas.	Study the trade-off between pulse shaping and occupied bandwidth.
`Spectral spacing` f_b/B_f	1.1–2.5; 0.1	1.6	Ratio $\rho = f_b/B_f$. It determines f_b and $f_a = f_b/2$. The lower limit implements the separation requirement $f_b \geq 1.1B_f$.	Explore spectral margin, harmonic frequencies, and switching periods.

Control	Range; step	Paper	Meaning and effect	Purpose
`Update factor` ζ	1–8; 1	2	Integer number of T_b periods for which each delay state remains active: $T_{\mathrm{PR}} = \zeta T_b$. A larger value slows key updating; a smaller one accelerates it.	Compare scrambling rates and synchronization requirements.
`Peak SNR`	5–35 dB; 1 dB	25 dB	Reference SNR at the intended maximum. It changes the noise floor, SINR curves, BER, and C_s; it does not change the noise-free array factors.	Study link robustness and secrecy capacity versus noise.

5.3 `Hardware realism` group

Control	Range; step	Paper	Interpretation
`Delay resolution`	0–250 ps; 25 ps	0 ps	At 0, ideal continuous delays are used. A nonzero value rounds each $D_n^1, \ldots, D_n^4$ to the nearest multiple of that resolution. This makes it possible to estimate null filling and SINR degradation caused by quantization.

Quantization is an extension provided by the demonstrator; the paper assumes ideal delays. In the benchmark, 250 ps is small compared with $T_b = 0.5\,\mu$s, so the change may be subtle. It becomes easier to observe by first increasing the bit rate or f_b/B_f.

5.4 Restoring the benchmark

The `Restore paper benchmark` button and the circular ↻ icon perform the same action. They restore the main parameters, the Σ receiver, the probe angle, and the first PR state.

Reproducibility detail for version 1.0

The restore button does not change the `PR seed` field, the selected harmonic, the `Pattern/SINR` mode, or the active tab. To reproduce the reference case exactly, also enter `PR seed = 2606` and select $\omega^{1,1}$ whenever metrics for the first harmonic are required.

6 Top strip of derived quantities

These boxes are not controls; they are results that update immediately:

Box	Calculation	How to interpret it
`Occupied bandwidth` B_f	$R_b(1+\alpha)/2$	Occupied bandwidth of the pulse-shaped QPSK signal.
`Dominant harmonic` $\omega^{-1,1}$	$f_c - f_a + f_b$	Absolute frequency of the beam governed by θ_2.
`Dominant harmonic` $\omega^{1,1}$	$f_c + f_a + f_b$	Absolute frequency of the beam governed by θ_1.
`Delay update` T_{PR}	ζ/f_b	Time between two consecutive delay/key sets.
`Time-modulation efficiency`	66%	Ideal fraction of radiated harmonic power in the two useful harmonics.

The 66% value is a property of the selected pulse waveforms and remains fixed in this version. At the receiver, the effective efficiency is halved because only one of the complementary Σ or Δ contributions is recovered.

7 Tab 01: `Beams & spectrum`

7.1 Harmonic selector

The $\omega^{-1,1} \cdot \theta_2$ and $\omega^{1,1} \cdot \theta_1$ buttons select the harmonic used for:

- the `Selected harmonic` and `Commanded pointing` boxes;
- γ_b, γ_e, and C_s;
- the exported CSV data;
- the encrypted-link tab and its constellations.

The selector does not hide the other family in the plot: both remain visible, but the active harmonic is drawn with greater emphasis.

7.2 Array diagram and architecture of the nth element

The left side of the diagram represents Eq. (1):

- the signal $s(t)$ is split into two branches;
- the upper branch uses D_n^1 with $a(t)$ and D_n^3 with $b(t)$;
- the lower branch uses D_n^2 with $a(t)$ and D_n^4 with $b(t)$;
- both branches are weighted by $1/\sqrt{2}$ and added to form $h_n(t)$;
- the PR controller updates all four delays.

The right side shows the $2N$ elements and two straight arrows: cyan for $\omega^{1,1}/\theta_1$ and orange for $\omega^{-1,1}/\theta_2$. The side boxes report the active harmonic, its pointing direction, and the physical aperture

$$L = (2N-1)d\lambda_c. \tag{9}$$

7.3 `Pattern` and `SINR` buttons

`Pattern` Displays $10\log_{10}|F(\theta)|^2$. Solid curves are Σ patterns and dashed curves are their complementary Δ patterns. Each Σ curve has a maximum at the commanded angle and the corresponding Δ curve has a null there.

`SINR` Displays recoverability with the Σ key versus angle for both harmonics. This is the most direct view for locating reception windows and regions that are unfavorable for an unintended receiver.

The vertical lines mark θ_1, θ_2, and `Eve*`. The latter is the point with the highest Σ SINR outside the excluded $\pm 8^\circ$ sector around the active beam; it does not represent a fixed position.

7.4 Metric boxes

Metric	Meaning
`Intended-direction SINR` γ_b	Σ-key recovery SINR exactly at the commanded angle of the active harmonic.
`Strongest unintended SINR` γ_e	Highest Σ SINR found outside the main $\pm 8^\circ$ sector. The note below shows $\gamma_b - \gamma_e$.
`Secrecy capacity` C_s	Value of Eq. (8), expressed in bit/s/Hz. If $\gamma_e \geq \gamma_b$, it is clipped to zero.

These three metrics always correspond to the selected harmonic, even though the angular plot simultaneously displays both families.

7.5 Spectrum and efficiency

The `Separated dominant replicas` panel represents the $q = 1$ slice for odd indices $i = -7, -5, -3, -1, 1, 3, 5, 7$. Beneath each bar, the offset $if_a + f_b$ relative to f_c is shown.

- The $i = \pm 1$ bars are the useful harmonics.
- The nearest unused component, $|i| = 3$, is $9.54\,\mathrm{dB}$ below $|i| = 1$.
- The strongest indicated degenerate component is $26.4\,\mathrm{dB}$ below.
- The two desired harmonics contain 66% of the ideal radiated harmonic power.

8 Tab 02: `Encrypted link`

This tab makes it possible to compare the received signal before descrambling with the recovery obtained by different receivers. The 180 QPSK observations are deterministic for the same parameter set, seed, and angle.

8.1 Receiver buttons

Table 3: Receiver presets.

Button	Automatic action	Interpretation
`Intended ů Σ key`	Sets `Probe angle` to θ_1 or θ_2, depending on the harmonic.	Intended receiver with the Σ key. This corresponds to the clean-recovery case in Fig. 4(a): Σ maximum and Δ null.
`Off-axis ů Σ key`	Sets the probe to the `Eve*` angle.	Stress case from Fig. 4(b). The receiver knows the Σ key, but spatial Σ/Δ interference prevents a clean constellation. This demonstrates that knowing the sequence alone is not sufficient in every direction.
`Authorized ů Δ key`	Sets the probe to the Δ-SINR maximum.	Authorized receiver outside the conventional beam. It uses the Δ key and exploits a complementary spatial window.
`No key`	Sets the probe to `Eve*` and omits descrambling.	Represents a receiver that does not know the PR sequence. The phase remains randomized and the BER is shown as $\simeq 0.5$.

After pressing a preset, you can manually move `Probe angle`; the selected receiver/key type is retained.

8.2 Security controls

Table 4: Controls in the encrypted-link tab.

Control	Range; step	Paper	Effect	Recommended use
`Probe angle`	0–180°; 0.5°	110° for $\omega^{1,1}$	Angle at which the two probe SINRs and the constellations are computed. It does not move the beams; it only moves the observation point.	Manually scan the angular domain and locate reception windows.
`PR phase states M`	2–100; 1	50	Number of equiprobable phases in Eq. (5). It changes the sequence, instantaneous phases, and delays, but not the ideal pattern envelope with continuous delays.	Compare small and dense PR alphabets.
`Timing mismatch`	0–100% of T_{PR}; 2%	0%	Normalized timing mismatch between transmitter and receiver. A fraction of the samples is compensated using the adjacent state, and the BER estimate includes the residual power.	Evaluate sensitivity to synchronization.
`PR seed`	1–999999; numeric field	2606	Seed of the deterministic pseudorandom generator. The same seed reproduces the same sequence; a different seed produces a different realization.	Repeatability, realization comparison, and scenario documentation.

8.3 `Next key state` button

It increments the PR-state index from #1 through #96 and then returns to the first state. ϕ_Σ, ϕ_Δ, and the associated delays change. With continuous delays, the phases change without shifting the maxima and nulls of the ideal patterns; with quantization, small variations may appear.

8.4 Data boxes and constellations

Box	Meaning
`Probe Σ-key SINR`	SINR obtained by a Σ receiver at `Probe angle`.
`Probe Δ-key SINR`	SINR obtained by a Δ receiver at the same angle.
`Selected-receiver BER`	BER of the selected receiver, including timing mismatch. In `No key`, it is fixed at $\simeq 0.5$.

The left constellation, `Before descrambling`, shows the superposition $\Sigma + \Delta$ before the key is applied. The right one shows the output after the selected processing. The four ideal QPSK symbol centers are marked as a reference. A compact cloud around them indicates recovery; a dispersed, rotated, or elongated cloud indicates noise, an incorrect key, or spatial interference.

8.5 Timing sensitivity

The lower panel applies the relationship from the paper

$$\frac{P_{\text{res}}}{P_{\text{useful}}} = 2\frac{\Delta t}{T_{\text{PR}}}. \tag{10}$$

For example, a 10% error produces a residual-power ratio of 0.20; a 25% error, 0.50; and a 50% error, 1.00. Degradation is gradual rather than an immediate binary failure.

9 Tab 03: `Delay synthesis`

9.1 Architecture diagram

The flow summarizes the implementation:

RF input $s(t)$ $\longrightarrow$ $1{:}2N$ distribution $\longrightarrow$ $2N$ cells with $D_n^1, \ldots, D_n^4$ $\longrightarrow$ Radiation $\omega^{-1,1} + \omega^{1,1}$.

The controller receives θ_1, θ_2, ϕ_Σ, and ϕ_Δ, solves the four delays for each cell, and holds them constant for T_{PR}.

9.2 `Recompute PR state` button

It has the same effect on the PR index as `Next key state`: it advances to the next state in the sequence and recomputes the table. The name emphasizes that, on this tab, the focus is on the new physical delay values.

9.3 Phase and period summary

The four boxes above the table show:

- ϕ_Σ and ϕ_Δ for the current state, in degrees;
- $T_a = 1/f_a$, the period of the modulation signal $a(t)$;

- $T_b = 1/f_b$, the period of the modulation signal $b(t)$.

9.4 Delay table

Each row corresponds to an element n:

Column	Content
`Element n`	Element index, from 1 to $2N$.
z_n/λ_c	Normalized position relative to the array center: $z_n/\lambda_c = [n - (2N+1)/2]d$.
D_n^1	Delay of $a(t)$ in the Σ branch; wrapped modulo T_a.
D_n^2	Delay of $a(t)$ in the Δ branch; wrapped modulo T_a.
D_n^3	Delay of $b(t)$ in the Σ branch; wrapped modulo T_b.
D_n^4	Delay of $b(t)$ in the Δ branch; wrapped modulo T_b.

The values are expressed in nanoseconds and represent a modulo-period solution. Therefore, two states may produce numerically different delays while yielding equivalent harmonic phases. If `Delay resolution` is nonzero, rounding is applied before the wrapped solution is displayed.

The final `Paper traceability` block contains no controls. It relates the architecture, delay synthesis, scrambling, and results to the equation and figure groups in the paper.

10 Global buttons and exported files

10.1 `Export angular data`

Downloads a CSV file for the currently selected harmonic. It contains 361 rows, from $0°$ to $180°$ in $0.5°$ steps, and the columns:

```
theta_deg,sum_pattern_db,difference_pattern_db,sigma_sinr_db,delta_sinr_db
```

The file is named according to the harmonic, for example `encrypted-dm-tma-omega-1-1.csv`. It is suitable for MATLAB, Python, Excel, or another plotting and comparison program.

10.2 `Save scenario`

Downloads `encrypted-dm-tma-scenario.json`. It stores:

- the selected harmonic;
- all global parameters, including the seed and resolution;
- B_f, f_a, f_b, f_b/B_f, the useful frequencies, T_{PR}, and the aperture;
- γ_b, γ_e, its angle, the best Δ window, and C_s;
- the model assumptions.

The JSON is a record, not a loadable file

Version 1.0 does not include a `Load scenario` button. The JSON is intended to document, archive, or externally process the scenario. To reproduce it in the interface, manually enter the values in its `parameters` block. Files are normally saved in the browser's Downloads folder.

11 Reference case from the paper

To reproduce the benchmark:

1. Press `Restore paper benchmark`.
2. In `Encrypted link`, enter `PR seed = 2606`.
3. In `Beams & spectrum`, select $\omega^{1,1} \cdot \theta_1$.
4. Select `Pattern` or `SINR`, depending on the quantity you want to view.

Parameter	Value	Associated result
$2N$; d/λ_c	20; 0.5	Normalized aperture $9.5\lambda_c$.
θ_1; θ_2	$110°$; $60°$	Independent cyan and orange beams.
f_c; R_b; α	$2.4\,\mathrm{GHz}$; $2\,\mathrm{Mbit/s}$; 0.25	$B_f = 1.25\,\mathrm{MHz}$.
f_b/B_f; ζ	1.6; 2	$f_a = 1\,\mathrm{MHz}$, $f_b = 2\,\mathrm{MHz}$, and $T_{\mathrm{PR}} = 1\,\mu\mathrm{s}$.
$f^{-1,1}$; $f^{1,1}$	$2.401\,\mathrm{GHz}$; $2.403\,\mathrm{GHz}$	Two spectrally separated useful replicas.
SNR; M	$25\,\mathrm{dB}$; 50	Security benchmark from the paper.
Resolution; timing error	$0\,\mathrm{ps}$; 0%	Ideal delays and synchronized receiver.

For $\omega^{1,1}$, the approximate results are

$$\gamma_b = 25.0\ \mathrm{dB}, \qquad \gamma_e = 4.4\ \mathrm{dB}, \qquad C_s = 6.39\ \mathrm{bit/s/Hz}, \tag{11}$$

consistent with the $\simeq 21\,\mathrm{dB}$ angular contrast and $C_s \simeq 6.5$ bit/s/Hz reported in the paper. Small differences may appear in the last displayed decimal because of rounding.

12 Guided experiments

Experiment 1 ů Independent beam steering

1. Restore the benchmark and open `Beams & spectrum`.
2. Move only f_+ `beam` θ_1 from $110°$ to $130°$.
3. Check that the cyan curves and arrow move, while the orange family remains at $60°$.
4. Return θ_1 to $110°$ and move θ_2.

Expected conclusion: each Σ/Δ pair is governed by its own angle and dominant harmonic.

Experiment 2 ů Distinguishing pattern from recoverability

1. With the benchmark loaded, alternate between `Pattern` and `SINR`.
2. In `Pattern`, identify Σ maxima and Δ nulls.

3. In `SINR`, compare these features with the recovery windows.

Expected conclusion: high radiated amplitude alone does not guarantee recovery; what matters is the relationship among the desired contribution, the complementary contribution, and noise.

Experiment 3 ů Knowing the key is not always enough

1. Select $\omega^{1,1}$ and open `Encrypted link`.
2. Press `Intended ů Σ key`: the processed constellation should cluster around the four QPSK symbols.
3. Press `Off-axis ů Σ key`: the key is still correct, but the cloud remains strongly affected by interference.
4. Press `No key`: the phase remains undescrambled and the BER is shown as $\simeq 0.5$.

Expected conclusion: security combines temporal scrambling and spatial Σ/Δ selectivity.

Experiment 4 ů Authorized reception outside the beam

1. In `Encrypted link`, press `Authorized ů Δ key`.
2. Observe the selected angle and compare `Probe Δ-key SINR` with `Probe Σ-key SINR`.
3. Slowly move `Probe angle` to either side of that position.

Expected conclusion: the Δ key opens authorized windows that do not necessarily coincide with the conventional Σ maximum.

Experiment 5 ů Synchronization error

1. Select `Intended ů Σ key`.
2. Compare `Timing mismatch` equal to 0, 10, 25, and 50%.
3. Record the $P_{\mathrm{res}}/P_{\mathrm{useful}}$ ratio, the BER, and the spread of the processed constellation.

Expected result: the residual ratios are 0, 0.20, 0.50, and 1.00, respectively, and degradation increases progressively.

Experiment 6 ů Number and spacing of elements

1. Keep $d = 0.5\lambda_c$ and compare $2N = 8$, 20, and 40.
2. Restore $2N = 20$ and compare $d = 0.25$, 0.50, and $0.75\lambda_c$.

Expected result: increasing the aperture narrows the beam; large spacings may introduce ambiguous maxima, especially far from broadside.

Experiment 7 ů Delay quantization

1. Set `QPSK bit rate` to $20\,\mathrm{Mbit/s}$ to increase the modulation frequencies and make the effect more visible.
2. Compare `Delay resolution` equal to 0, 100, and $250\,\mathrm{ps}$.
3. Observe the delay table, the Δ nulls, and the SINR metrics.

Expected result: the delays align with the timing grid; phase errors can partially fill the nulls and

reduce contrast.

Experiment 8 ů Frequency scaling

1. Export or note the pattern with $f_c = 2.4\,\mathrm{GHz}$.
2. Change only the carrier to $10\,\mathrm{GHz}$.
3. Compare the pattern, absolute frequencies, and aperture in centimeters.

Expected result: the normalized pattern is preserved, while λ_c and the physical aperture decrease and the replicas shift in frequency.

13 What changes when each parameter is modified

The following table is a quick reference. “Yes” means a directly visible effect; “indirect” means that the change appears through a derived quantity or the PR realization.

Parameter	Angular pattern	Frequencies	Delays	SINR/C_s	Constellation	Note
$2N$	**Yes**	No	**Yes**	**Yes**	**Yes**	Changes aperture and coherent summation.
d/λ_c	**Yes**	No	**Yes**	**Yes**	**Yes**	May create grating lobes.
θ_1, θ_2	**Yes**	No	**Yes**	**Yes**	**Yes**	Each angle governs one family.
f_c	No, normalized	**Yes**	No, in this normalized model	No	No	Changes λ_c and physical size.
$R_b, \alpha, f_b/B_f$	Ideally no	**Yes**	**Yes**	Indirect with quantization	Indirect	Scale timing and spectral separation.
ζ	No	No	No	Only with timing error	**Yes**	Changes T_{PR}.
SNR	No	No	No	**Yes**	**Yes**	Changes noise and BER.
Delay resolution	**Yes**	No	**Yes**	**Yes**	**Yes**	Optional quantization.
M and seed	Ideally no	No	**Yes**	Indirect	**Yes**	Change the PR realization.
Probe angle	No	No	No	Local reading	**Yes**	Does not move the beam.
Timing error	No	No	No	Effective BER	**Yes**	Does not change the baseline spatial SINR.

14 Assumptions, scope, and precautions

The demonstrator is an analytical, array-factor-level model. It assumes:

- isotropic elements with uniform excitation;
- a coherent channel and, except for the explicit control, perfect synchronization;
- ideal switches and continuous delays, except when optional quantization is enabled;

- no mutual coupling, RF losses, component dispersion, amplitude/phase errors, phase noise, or nonlinearities;
- a LOS-type angular scenario, without an explicit multipath channel;
- ideal spectral isolation of the analyzed harmonic through filtering.

Therefore, the results can be used to verify the architecture, explore trends, and prepare specifications, but they do not replace a full-wave electromagnetic simulation, an RF circuit model, or hardware measurements.

Good practice when sharing results

Store the CSV, JSON, and a note with the demonstrator version together. Always state the selected harmonic and whether delay quantization was used. For a reproducible comparison, also record M, the seed, the PR state, and the probe angle.

15 Frequently asked questions

Because the selector sets the detailed metrics, constellations, and export; the plot retains both families to show their independence.

Because the geometry is expressed in λ_c. The physical dimensions and absolute frequencies change, but the normalized array factor does not.

Outside the appropriate direction, the complementary contribution may dominate. The key removes the PR phase, but it does not remove spatial Σ/Δ interference.

This is the random-decision assumption when the scrambling sequence is unknown; it is not an estimate obtained from valid coherent demodulation.

The common pseudorandom phase changes between intervals, while the relative phase laws preserve the commanded maximum and null. Quantization may slightly break this invariance.

Not in version 1.0. The JSON documents the scenario; its parameters must be re-entered manually.

Export follows the active harmonic. Select and export each one separately if you need to compare both.

A Checklist for a public demonstration

☐ Extract the package and open `index.html` before the session.

☐ Press `Restore paper benchmark` and set `PR seed = 2606`.

☐ Show the independence of θ_1 and θ_2.

☐ Alternate between `Pattern` and `SINR`.

☐ Verify approximately 25.0 dB, 4.4 dB, and 6.39 bit/s/Hz for $\omega^{1,1}$.

☐ Compare the four receiver modes.

☐ Show a PR-state change and the $D_n^1, \ldots, D_n^4$ table.

☐ Export an example CSV and JSON file.

☐ Remind the audience of the assumptions of the analytical model.

B Benchmark quick reference

$2N$	20	d/λ_c	0.5
θ_1	110°	θ_2	60°
f_c	$2.4\,\mathrm{GHz}$	R_b	$2\,\mathrm{Mbit/s}$
α	0.25	B_f	$1.25\,\mathrm{MHz}$
f_a	$1\,\mathrm{MHz}$	f_b	$2\,\mathrm{MHz}$
f_b/B_f	1.6	ζ	2
T_{PR}	$1\,\mu\mathrm{s}$	M	50
Seed	2606	SNR	$25\,\mathrm{dB}$
Resolution	$0\,\mathrm{ps}$	Timing error	0%